

1 Vanadium, a transition metal, is commercially important.

(a) (i) Write down the electronic configuration of vanadium, and hence, write down the electronic configuration of V^{2+} ion.

(ii) Explain why $V^{2+}(aq)$ is coloured whereas $Ca^{2+}(aq)$ is colourless.

[4]

(b) Sketch the general trends of the first ionisation energy and the third ionisation energy of the elements from Ca to Cu on the **same** axes. Explain the trends shown for these two properties.

[5]

(c) Both Ca and Zn play an important role biologically in the body system. Vanadium ions are poisonous as they can cause bone porosity by replacing the calcium ions in bones. One treatment for vanadium poisoning involves administering a solution of edta, which forms hexacoordinate complexes with many divalent metal ions using the 4 donor oxygen and two donor nitrogen atoms.

Comment on the use of edta as a treatment for vanadium poisoning in the light of the following data:

Equilibrium					$K_c / \text{mol}^{-1} \text{ dm}^3$
Ca^{2+}	+	$edta^{4-}$	$\rightleftharpoons$	$[Ca(edta)]^{2-}$	5×10^{10}
Zn^{2+}	+	$edta^{4-}$	$\rightleftharpoons$	$[Zn(edta)]^{2-}$	3×10^{16}
V^{2+}	+	$edta^{4-}$	$\rightleftharpoons$	$[V(edta)]^{2-}$	4×10^{16}

What problem might arise during the treatment and how could it be overcome?

[2]

(d) Transition metals are often good homogeneous and heterogeneous catalysts.

V_2O_5 is used as catalyst to speed up the conversion of SO_2 into SO_3 in the Contact process for making sulfuric acid.

The key reaction in the Contact process is as follows:



(i) For the reaction above, identify the type of catalyst involved and explain clearly how the catalyst works.

When a 2:1 ratio of sulfur dioxide and oxygen at a total initial pressure of 3 atm is passed over V_2O_5 , the catalyst in a fixed volume steel vessel at 430°C , the partial pressure of sulfur trioxide at equilibrium is found to be 1.9 atm.

- (ii) Calculate the percentage conversion of SO_2 to SO_3 and the equilibrium constant, K_p at 430°C .
 - (iii) According to Le Chatelier's Principle, the reaction will be favoured by a low temperature and a high pressure. Give one reason (other than cost factor) why a higher pressure is not applied.
 - (iv) Comment on the effect on the percentage conversion if another unreactive but toxic gas was accidentally added to V_2O_5 .
- [6]

- (e) Given the following information on the colours of the aqueous vanadium ion of various oxidation states:

VO_2^+ (yellow) VO^{2+} (blue) V^{3+} (green) V^{2+} (violet)

Use relevant E^θ values from the *Data Booklet* to predict the colour observed upon mixing acidified VO_2^+ solution with excess aqueous Sn^{2+} solution. Calculate the relevant E^θ_{cell} values and write balanced equations for reactions that occur.

[3]

[Total: 20]

- 2 (a) The solubility products of some group II compounds at 25°C are shown in the following table.

Compound	K_{sp}
$\text{Ba}(\text{OH})_2$	5.0×10^{-3}
$\text{Ca}(\text{OH})_2$	?
BaCO_3	5.1×10^{-9}
CaCO_3	3.8×10^{-9}

With reference to the given data,

- (i) Write an expression for the solubility product of $\text{Ca}(\text{OH})_2$.
- (ii) Calculate the pH of a saturated solution of $\text{Ca}(\text{OH})_2$ at 25°C , given that its solubility is 0.830 g dm^{-3} .
- (iii) Determine the solubility product of $\text{Ca}(\text{OH})_2$ at 25°C , stating its units.

- (iv) Predict qualitatively the effect (if any) on the solubility and solubility product of $\text{Ca}(\text{OH})_2$ when 25.0 cm^3 of $0.050 \text{ mol dm}^{-3}$ solution of potassium hydroxide is added to the solution in (ii).

Barium hydroxide is also known as baryta water and just like limewater, may be used to detect the presence of carbon dioxide through the formation of a white precipitate.

- (v) Write a balanced equation to show the reaction between carbon dioxide and baryta water.
- (vi) Which solution, limewater or baryta water, is more sensitive to the detection of carbon dioxide gas at 25°C ? Explain your answer with reference to the data given in the table above.

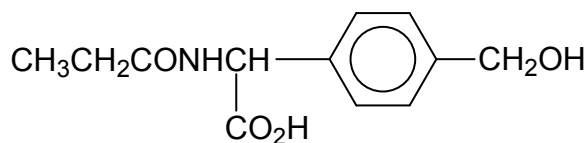
[9]

- (b) White lead, $\text{Pb}(\text{OH})_2 \cdot \text{PbCO}_3$, used to be the primary white pigment used in paints but has since been replaced, largely by barium sulfate and titanium(IV) oxide due to the toxicity of lead.

- (i) Assuming that lead(II) compounds react in the same way as group II compounds, write a balanced equation, with state symbols, to illustrate the action of heat on 'white lead'.
- (ii) Calculate the percentage loss in mass on heating a sample of 'white lead' until no further change occurred.
- (iii) Would you expect the temperature required for the decomposition of white lead to be higher or lower than that of $\text{Ca}(\text{OH})_2 \cdot \text{CaCO}_3$? Explain your answer.

[5]

- (c) Compound **P**, with the structure shown below, has two functional groups that may exhibit acidic properties.

**P**

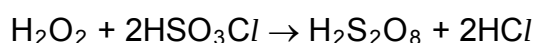
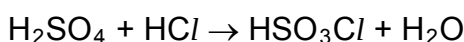
- (i) State and explain which of these two groups has a higher acid strength. Hence give the formula of the product formed when aqueous sodium carbonate is added to compound **P**.
- (ii) **P** can be synthesized by reacting an amino acid and an acyl chloride. Write an equation for its synthesis. An isomer of **P** was also formed in the reaction. Give the structure of this isomer.

[6]

3 This question is about Period 3 elements and their compounds.

- (a) A sample of magnesium oxide is contaminated with some aluminium oxide. By making use of the fact that Period 3 oxides have different acid-base properties, describe a simple method by which a pure sample of magnesium oxide can be obtained. Include an equation for any reaction that is utilized. [2]

- (b) Another Period 3 oxide, sulfur trioxide dissolves in water to form sulfuric acid, H_2SO_4 which can be converted into peroxodisulfuric acid, $\text{H}_2\text{S}_2\text{O}_8$ via the two-step process below, with chlorosulfonic acid, HSO_3Cl , as an intermediate:



- (i) The boiling point of H_2SO_4 is 290°C whereas that of HSO_3Cl is 152°C . By comparing the structures of the two compounds, explain the difference in the boiling points.
- (ii) Give the displayed formula of the $\text{H}_2\text{S}_2\text{O}_8$ molecule, showing the spatial arrangement and estimated bond angle around any one sulfur atom and one oxygen atom. [5]
- (c) Chlorine forms an oxide, Cl_2O_7 , with the structure $\text{O}_3\text{Cl}/\text{OC}/\text{O}_3$ and a boiling point of 82°C . Its standard enthalpy change of formation is -546 kJ mol^{-1} .
- (i) Write the equation which corresponds to the standard enthalpy change of formation of Cl_2O_7 .
- (ii) Given that the bond energy of $\text{Cl}-\text{O}$ bond is 269 kJ mol^{-1} and using relevant data from the *Data Booklet*, estimate the average bond energy of the $\text{Cl}=\text{O}$ bond.
- (iii) The bond energy calculated in (ii) could have been more accurate if the value of an additional enthalpy change is known. What process does this enthalpy change correspond to? [4]

- (d) (i) Define the term *lattice energy*.
- (ii) State and explain whether the oxide or fluoride ion is smaller.
- (iii) Refractories are heat-resistant compounds used to line furnaces. Which compound, magnesium oxide or magnesium fluoride, is more suitable for use as a refractory? Explain your answer. [5]

- (e) Magnesium fluoride, differs in its bonding from that of sulfur difluoride, SF_2 , as reflected by the difference in their physical properties.
- Explain why the difference in bonding arises.
 - Suggest one difference in physical properties that is likely to ensue from the difference in bonding.
 - The compound SF_2 readily reacts with fluorine to give SF_4 . Suggest why MgF_2 does not react with more fluorine to give MgF_4 .

[4]

[Total: 20]

- 4 Compound **P** is a sweet smelling substance extracted from a fern which grows in the tropical region. Acid hydrolysis of compound **P** yields three products, compounds **A**, **B** and **C**. Compound **C** was precipitated out as white solid and was filtered off the reaction mixture.



A series of tests were then performed on compound **A** and compound **B**.

- (a) A gaseous sample of 0.146 g of compound **A** took up a volume of 80.1 cm^3 at a temperature of 400 K and a pressure of $1.01 \times 10^5 \text{ Pa}$. In addition, effervescence of a colourless gas was observed when aqueous sodium carbonate is added to compound **A**.

Use these data to calculate the M_r of compound **A** and hence suggest a possible structure.

[2]

- (b) Reacting compound **A** and compound **B** with phosphorus pentachloride, PCl_5 , yield compound **D** and compound **E** respectively. Effervescence of white fumes was observed.

Both compound **D** and **E** were then hydrolysed with aqueous sodium hydroxide and the kinetics rate data were collected.

- (i) The results for the hydrolysis of compound **D** are as shown.

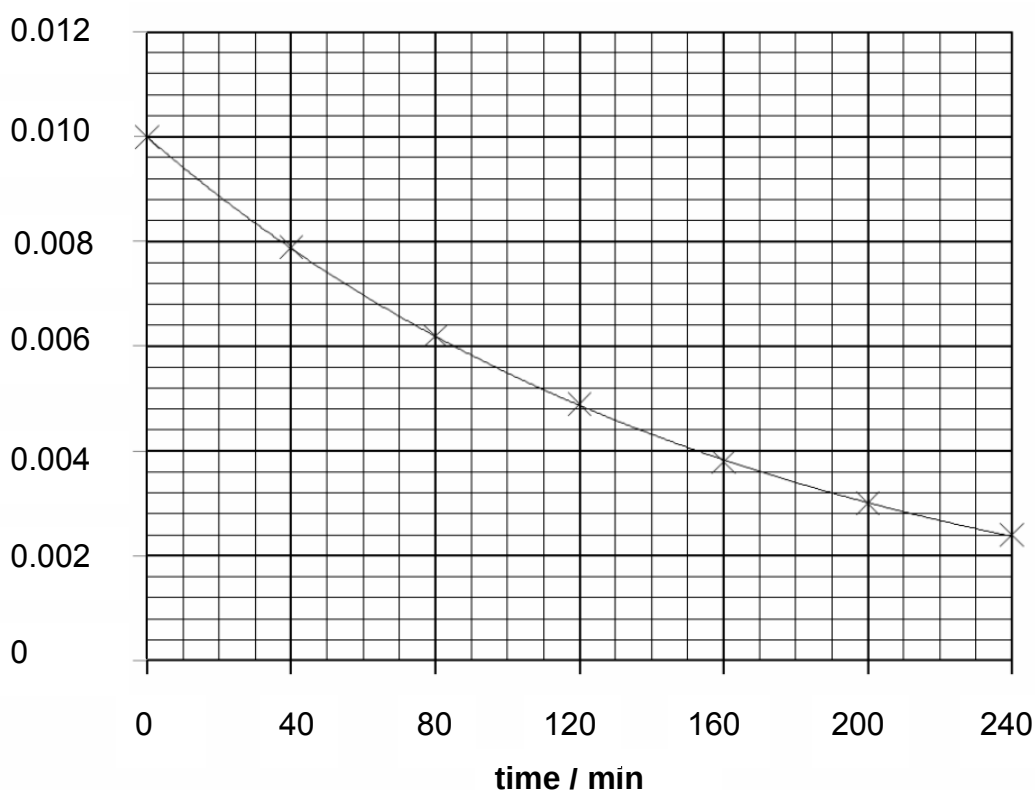
Experiment	[D] / mol dm^{-3}	[OH^-] / mol dm^{-3}	Initial rate / $\text{mol dm}^{-3} \text{s}^{-1}$
1	0.0100	0.120	1.8×10^{-4}
2	0.0050	0.240	1.8×10^{-4}
3	0.0050	0.480	3.6×10^{-4}

Using the above data, deduce the order of reaction with respect to compound **D** and OH^- .

- (ii) Hence, construct a rate equation and use it to calculate a value of the rate constant, k_1 , including its units, for the hydrolysis of compound **D**. [3]

- (c) The results for the hydrolysis of compound **E** are shown in the form of graphical representation. The overall concentration of aqueous NaOH remained virtually constant at $0.120 \text{ mol dm}^{-3}$.

[E] / mol dm⁻³



- (i) Use the graph above to determine the following:

- I Use the half life method to deduce the order of reaction with respect to compound **E**.
- II Given that the same plot was obtained when the experiment was repeated with $0.240 \text{ mol dm}^{-3}$ of aqueous sodium hydroxide, calculate the value of the rate constant, k_2 , including its units, for the hydrolysis of compound **E**.

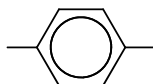
- (ii) Using the initial rate data given in **b(i)** and the graph above, calculate the ratio of the initial rate of the hydrolysis of compound **D** to compound **E**. Hence, suggest the functional group which is likely to be present in compound **B**. Explain clearly your reasoning.

[7]

- (d) Compound **B** has the molecular formula $C_9H_{12}O_2$ and is monobasic. Reacting compound **B** with aqueous bromine at room temperature produces compound **F** which is symmetrical. Compound **B** does not react with hot, acidified aqueous $Na_2Cr_2O_7$. Using the data given above, suggest the structural formulae of compound **B** and compound **F**.

[2]

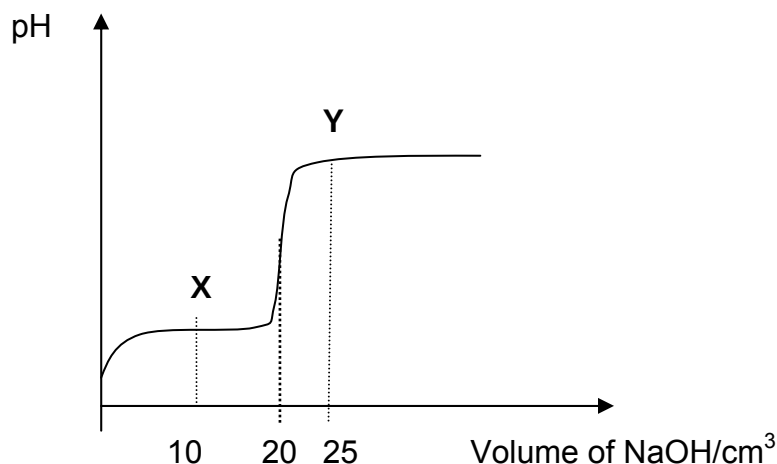
- (e) Given that compound **C** is identified to be 4-methylbenzoic acid.



Draw a possible structural formula of compound **P**.

[1]

- (f) Compound **A** can be converted to 2-chloroethanoic acid, $ClCH_2COOH$. 10.0 cm^3 of 0.100 mol dm^{-3} of 2-chloroethanoic acid is titrated against $0.0500\text{ mol dm}^{-3}$ of $NaOH$. The graph of pH against volume of $NaOH$ added is shown:



- (i) Given that the K_a of $ClCH_2COOH$ is $1.4 \times 10^{-3}\text{ mol dm}^{-3}$, calculate the initial pH.
- (ii) Calculate the pH at points **X** and **Y**.

[5]

[Total: 20]

- 5 Compounds containing nitrogen are important to life and have applications in science and medicine.

TNT (2, 4, 6 – trinitromethylbenzene), was first prepared in 1863 by German chemist Joseph Wilbrand and originally used as a yellow dye. Its potential as an explosive was not appreciated for several years mainly because it was so difficult to detonate and because it was less powerful than alternatives. TNT is quite an explosive material. It can quickly destroy anything within its blast radius when detonated.

In the process of detonation, TNT is oxidised in the presence of excess oxygen into the stable products of CO_2 , H_2O and N_2 . The energy produced during the formation of these products from 1 mole of TNT is known as enthalpy of detonation, ΔH_d , in kJ mol^{-1} .

- (a) (i) Write down a balanced equation showing the detonation of 1 mole of solid TNT. Using the data below, construct an energy cycle and use it to calculate the standard enthalpy change of detonation of TNT.

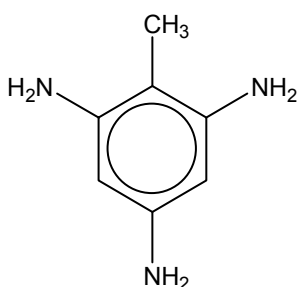
Standard enthalpy change of formation of TNT = -54 kJ mol^{-1}
 Standard enthalpy change of formation of H_2O = -242 kJ mol^{-1}
 Standard enthalpy change of combustion of carbon = -394 kJ mol^{-1}

- (ii) The energy obtained from experimentally detonating TNT was $-2545 \text{ kJ mol}^{-1}$. Suggest a reason for the discrepancy from the value that was obtained in (i).
- (iii) Deduce the sign of ΔS for the detonation of TNT. Hence, or otherwise, explain why the detonation process is always spontaneous.
- (iv) Explain, in energy terms, why TNT is so explosive.

[7]

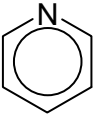
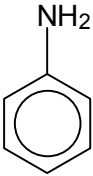
- (b) TNT can be synthesized in the laboratory in a three-step nitration of methylbenzene.

- (i) Name and describe the mechanism of the **first** step of the synthesis of TNT from methylbenzene.
- (ii) "Subsequent nitration of the ring after the first one becomes more difficult". Comment on this statement.
- (iii) TNT can be converted into a non-toxic form by reducing it to the amine as shown.



State the reagent(s) and conditions required for this conversion. [5]

(c) The pK_b values of some nitrogeneous bases are given below.

Methylamine	Trimethylamine	Ammonia	Hydroxylamine	Pyridine	Phenylamine
CH_3NH_2	$(\text{CH}_3)_3\text{N}$	NH_3	NH_2OH		
3.34	4.19	4.74	6.80	8.75	9.37

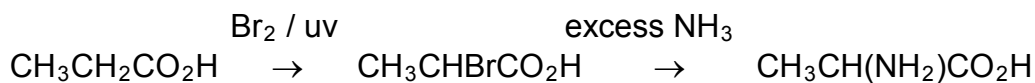
Explain the relative basicity of

- (i) methylamine and trimethylamine;
- (ii) ammonia and hydroxylamine;
- (iii) pyridine and phenylamine.

[4]

(d) Explain the following observations:

- (i) Alanine, $\text{CH}_3\text{CH}(\text{NH}_2)\text{CO}_2\text{H}$, extracted from silkworm is optically active whereas alanine synthesised via the following reaction scheme is optically inactive.



- (ii) Egg white (which is originally colourless and transparent) turns white and opaque when an aqueous solution containing heavy metal ions such as Cu^{2+} is added to it.

[4]

[Total: 20]